

Experiment 6: Implementation of a 3x8 Decoder.

Theory: A 3×8 decoder is a combinational logic circuit that converts 3 binary input lines into 8 unique output lines. For each combination of inputs, only one output becomes HIGH (1), while all others remain LOW (0). It is also known as a one-hot output device.

3×8 decoder typically has:

3 inputs: A, B, C

8 outputs: Y₀ to Y₇

Enable input (E): Controls whether the decoder is active

When the enable input is HIGH, the decoder functions normally. When it is LOW, all outputs remain LOW regardless of input values.

3 to 8 Line Decoder

Inputs			Outputs							
x	y	z	D0	D1	D2	D3	D4	D5	D6	D7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

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Boolean Expressions: (Let us think xyz=abc)

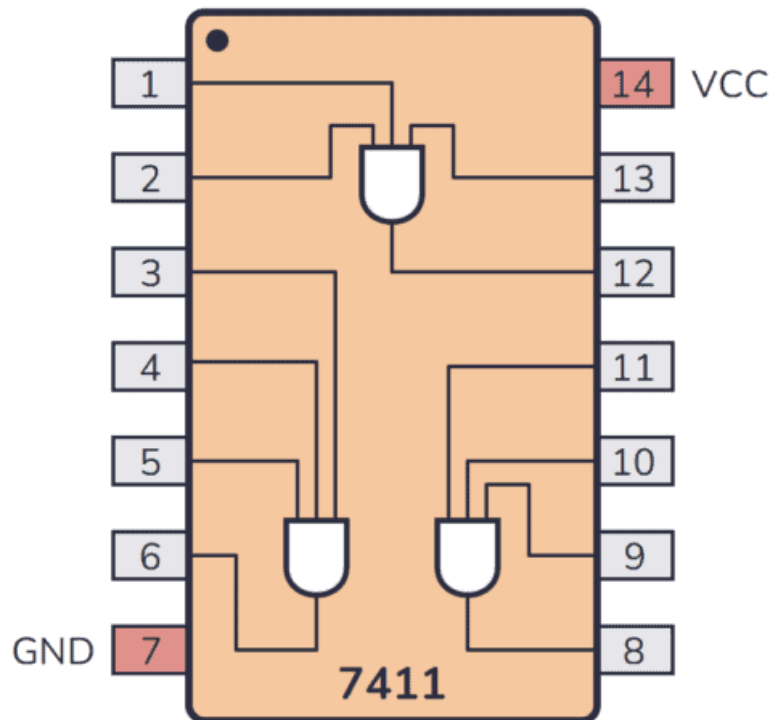
- $Y_0 = A'B'C'$
- $Y_1 = A'B'C$
- $Y_2 = A'BC'$
- $Y_3 = A'BC$
- $Y_4 = AB'C'$
- $Y_5 = AB'C$
- $Y_6 = ABC'$
- $Y_7 = ABC$

Apparatus Required:

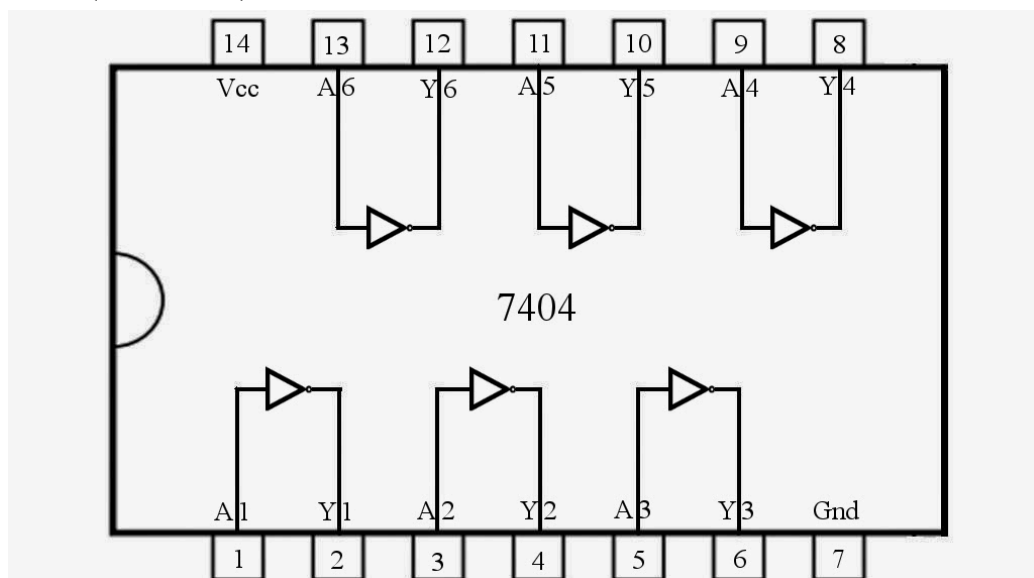
- Breadboard
- Connecting Wires
- IC 7411
- IC 7404 NOT Gate
- Power Supply (+5V)
- LEDs (for output indication)
- Resistors

IC Structure:

1. IC 7411 (3 Input AND Gate):



2. IC 7404 (NOT Gate):



Procedure:

1. Place the IC 74138 on the breadboard.
2. Connect the power supply (+5V and GND) to the IC.
3. Connect the enable pins appropriately (two LOW and one HIGH for activation).

4. Provide input signals A, B, and C using switches.
5. Connect LEDs to each output (Y_0 – Y_7) with resistors.
6. Apply different combinations of inputs.
7. Observe and record which output LED glows for each input.

Result: The outputs are observed according to the truth table, where only one output is HIGH for each input combination.

Conclusion: Thus, the 3×8 decoder is successfully implemented and verified. It correctly converts 3-bit binary inputs into one of eight outputs. This circuit is widely used in memory addressing, instruction decoding, and digital system design.